

CSE4227 Digital Image Processing

Chapter 11 Representation and Description (Chain Code)

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Today's Contents

- ❑ What is Image representation?
- ❑ What information we can use to represent an image?
- ❑ Boundary Based Image representation
- ❑ Chain Code
 - Types
 - Properties
 - Problems in Chain Code Representation
 - Solutions
 - Advantages & Disadvantages
 - Examples of Shape Number

■ Chapter 11 from R.C. Gonzalez and R.E. Woods, Digital Image Processing (3rd Edition), Prentice Hall, 2008 [**Section 11.1.1, 11.1.2**]

Image Representation

Objective:

To represent and describe information embedded in an image in **other forms that are more suitable** than the image itself.

Benefits:

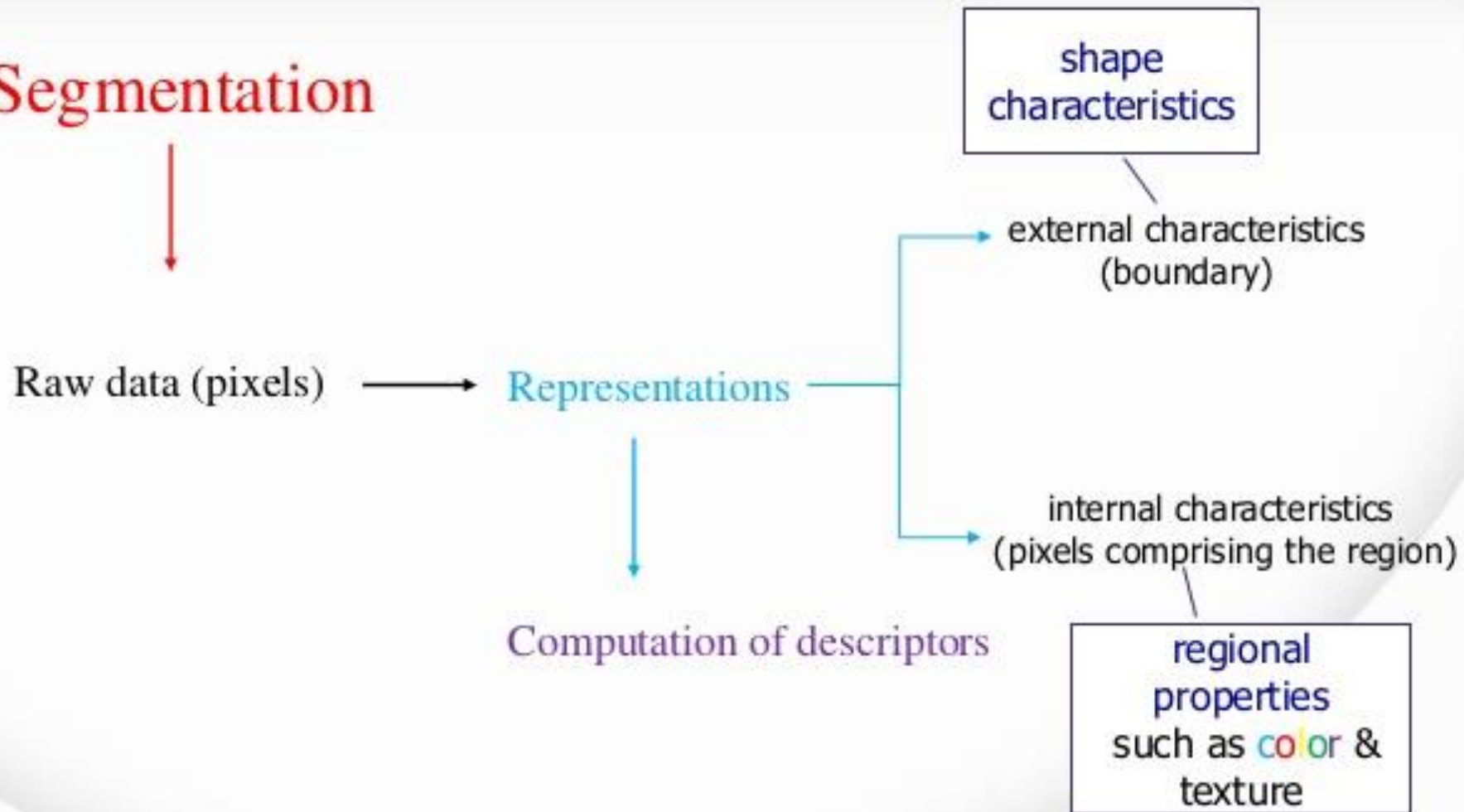
- Easier to understand
- Require fewer memory, faster to be processed
- More “ready to be used”

What kind of information we can use?

- Boundary, shape
- Region
- Texture
- Relation between regions

Image Processing

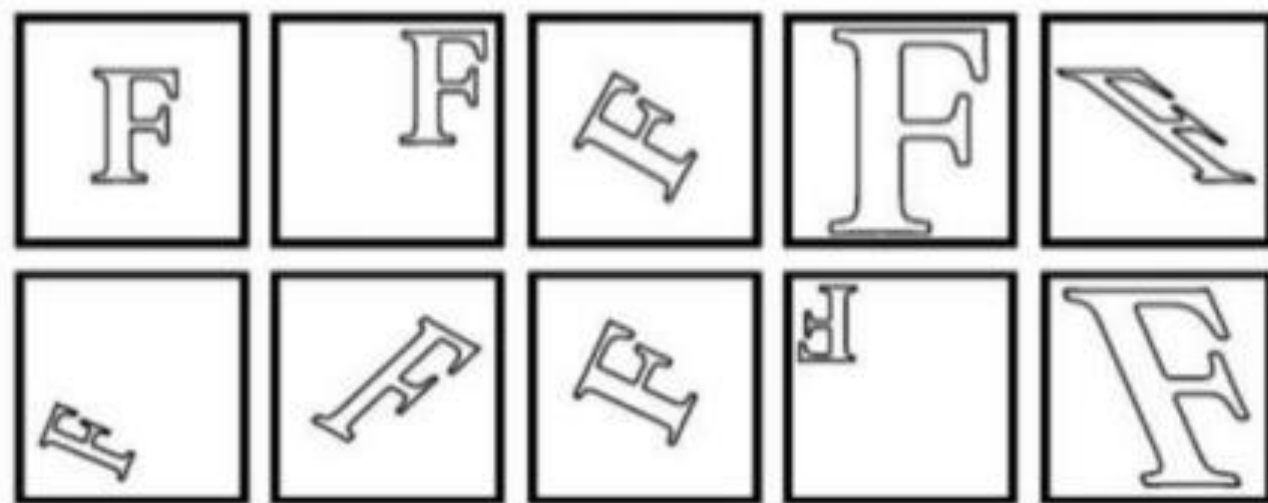
Segmentation



Representation and Description

- After segmenting an image to find regions of interest, we want to represent those regions in a concise form
- This can be used for
 - Recognition
 - Compression
 - Further processing (e.g., joining, simplifying, tracking)
- Two main approaches
 - Boundary based methods
 - Region based methods

Is invariance needed?



- Translation invariance
- Scale invariance
- Rotation invariance

Boundary based Image Representation

Chain Code

Freeman Chain codes are actually one of the oldest techniques in computer vision originally introduced.

The basic principle of **chain codes** is to **separately encode** each **connected component**, or "blob", in the **image**.

Introduction

- A technique to represent the region or object by describing its contour.
- Works on connected component
- A chain code is a boundary representation technique by which
“A contour is represented as a sequence of straight line segments of specified length (usually 1) and direction”.

Types of Chain Code

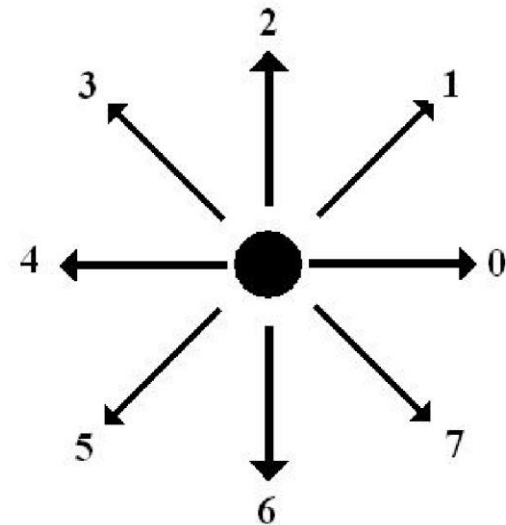
- 8-neighbor : It works with 8-neighbor of a contour pixel. It will follow the contour and keep generating direction number that is generated in anti-clockwise manner.
- 4-neighbor: This representation also known as crack code. Works with 4-neighbor of pixel which can define the boundary line.

Continue.....

- ❑ The codes associated with eight possible directions are the chain codes
- ❑ With x as the current contour pixel position, the codes are generally defined as:

Chain codes =

3	2	1
4	x	0
5	6	7



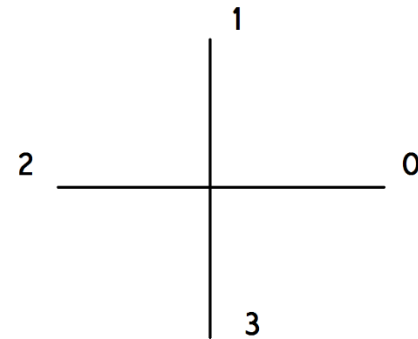
Properties of 8-neighbor Chain Code

- ❑ **Even codes** {0,2,4,6} correspond to **horizontal and vertical** directions; **odd** codes {1,3,5,7} correspond to the **diagonal** directions.
- ❑ Each code can be considered as the angular direction, in multiples of 45deg., that we must move to go from one contour pixel to the next.

4-neighbor Chain code

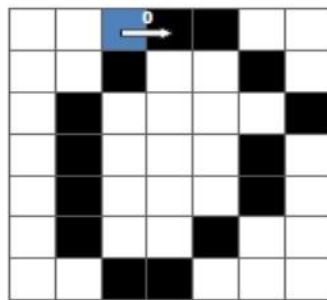
The "crack" code can be viewed as a chain code with four possible directions instead of eight.

Crack codes = $\begin{matrix} 1 \\ 2 & x & 0 \\ 3 \end{matrix}$

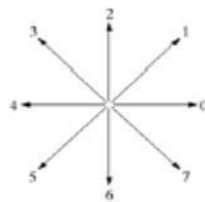


Steps of chain codes

- ❑ Select some starting point of the boundary and represent it by its absolute coordinates in the image
- ❑ Represent every consecutive point by a chain code showing transition needed to go from current point to next point on the boundary
- ❑ Stop if the next point is the *initial point* or the *end of the boundary*

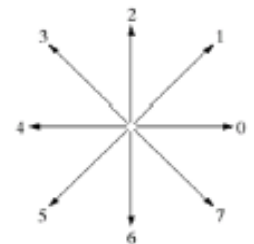
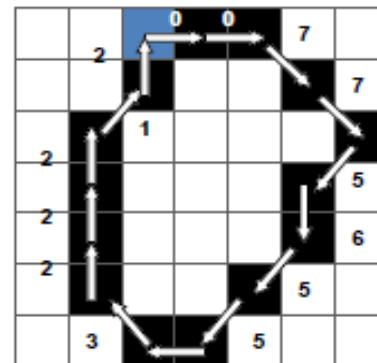


BOUNDARY



Chain Code:

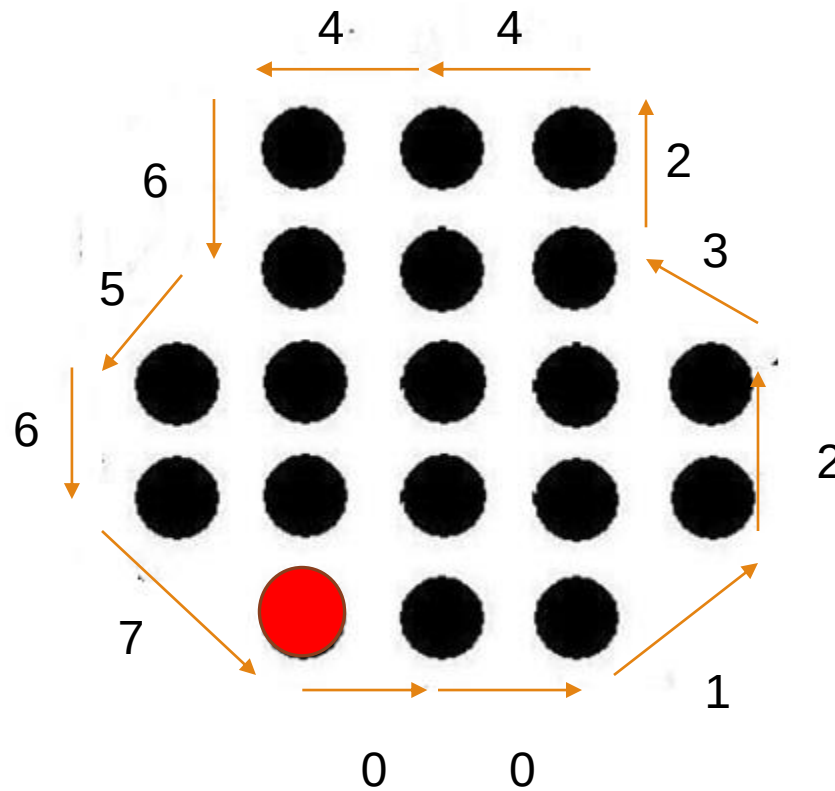
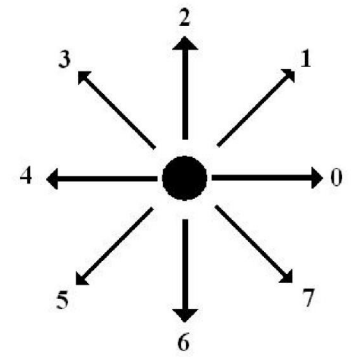
0



Chain Code:

0, 0, 7, 7, 5, 6, 5, 5
4, 3, 2, 2, 2, 1, 2

Simulation

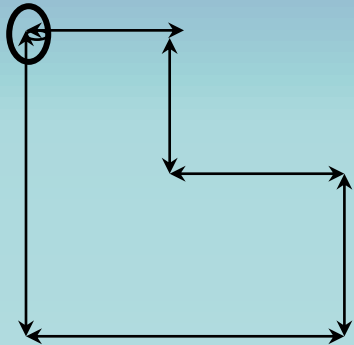


 = Starting

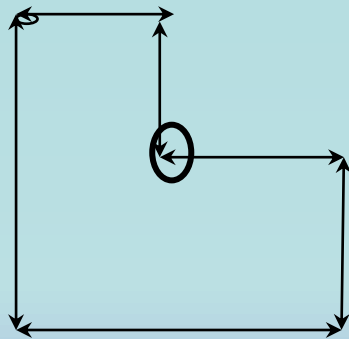
Advantages

- Boundary reduced to a sequence of numbers
- Provides good compression of boundary description
- They are translation invariant

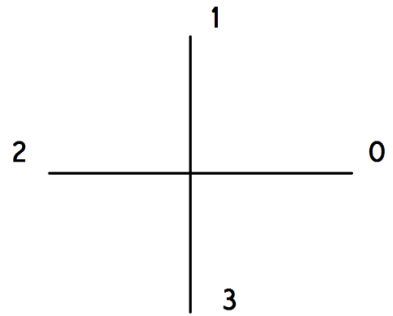
1) Change Starting Point



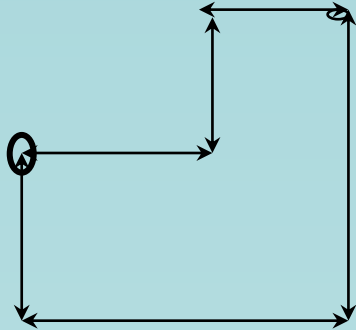
CC: 03032211



CC: 03221103

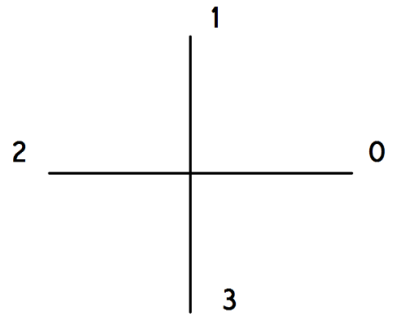


3) Reflex

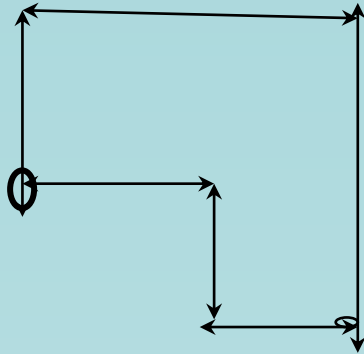


CC:

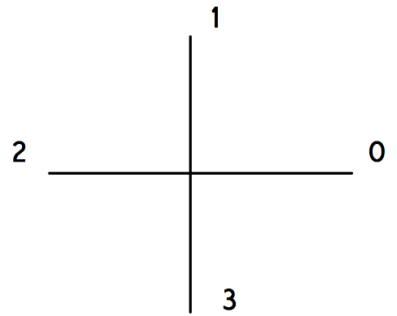
01033221



3) Rotation



CC: 10033212



Chain Code changes in

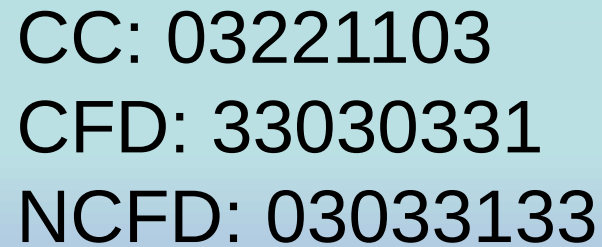
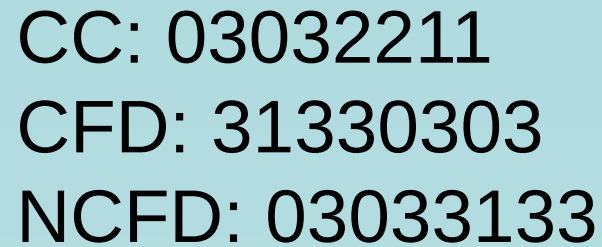
- 1. Changing start point**
- 2. Reflex**
- 3. Rotation**

❖ **Solution is Normalization**

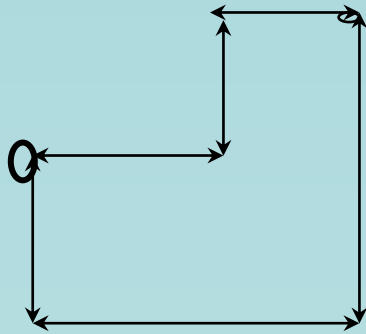
❖ **3 steps:**

- 1. Calculate chain code**
- 2. Calculate circular first difference (CFD)**
- 3. Calculate Normalize CFD (aka Shape Number)**

A coordinate system with a horizontal axis and a vertical axis intersecting at the origin. The horizontal axis has a label '0' at its right end and '2' at its left end. The vertical axis has a label '1' at its top end and '3' at its bottom end.



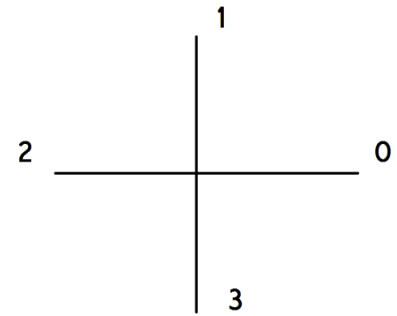
2) Reflex



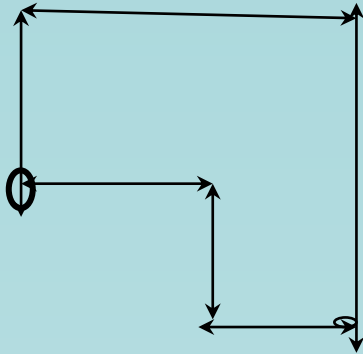
CC: 01033221

CFD: 13303033

NCFD: 03033133



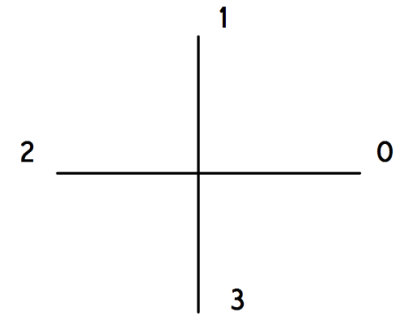
3) Rotation



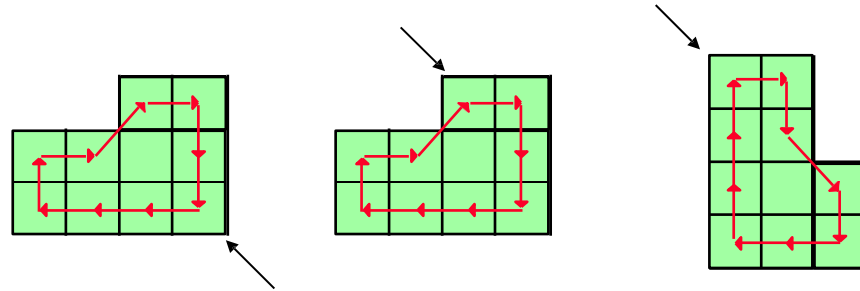
CC: 10033212

CFD: 30303313

NCFD: 03033133



Chain Code Independent of Location, Starting Point and Orientation



Chain Code:

444201066

066444201

067644222

Curvature = differences in chain code values :

0022-11-602

-6020022-11

-6-11202002

Normalize curvature = mod 8 :

002271202

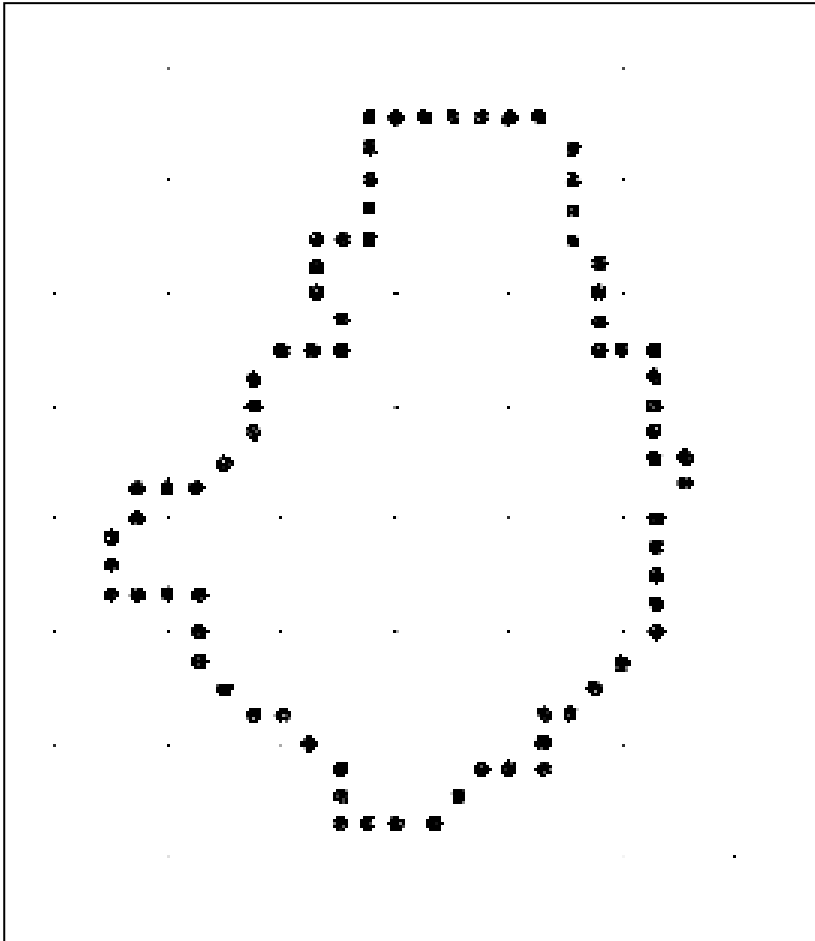
202002271

271202002

Code is the cyclic permutation which produces the smallest number:

002271202

Another Example:



Problem1:

- Long chains of codes
- Easily disturbed by noise, and sidetracked

Problem2:

- start point

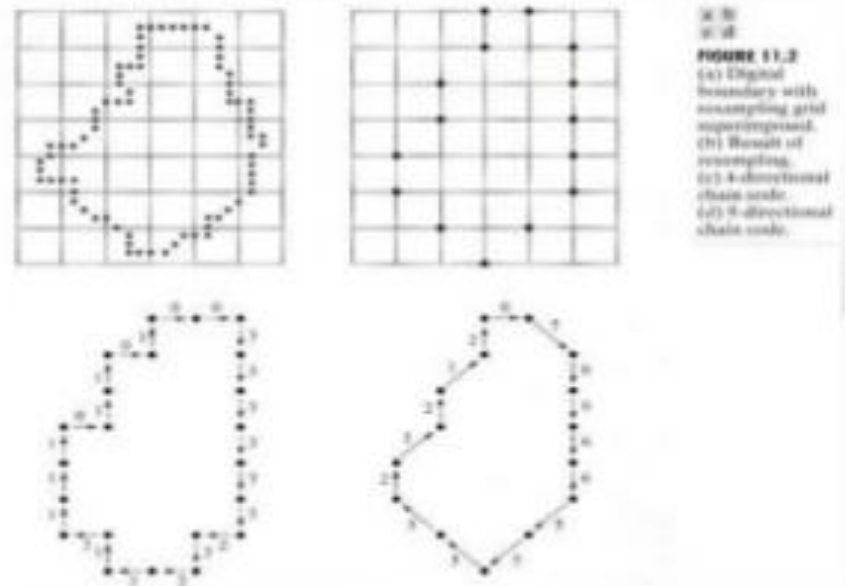
Chain Codes

Problem1:

- Long chains of codes
- Easily disturbed by noise, and sidetracked

Solution:

- Resampling using larger grid spacing

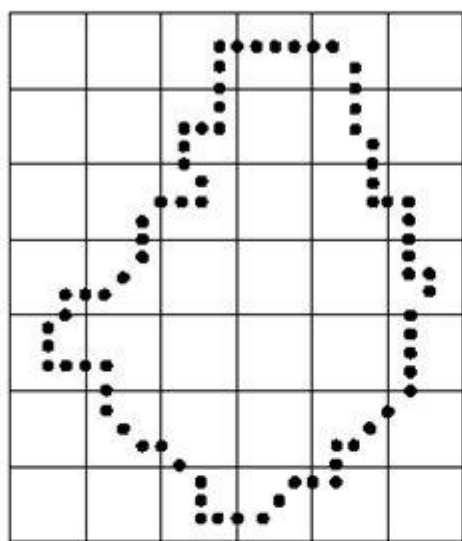


Before resample: 000000766...1111

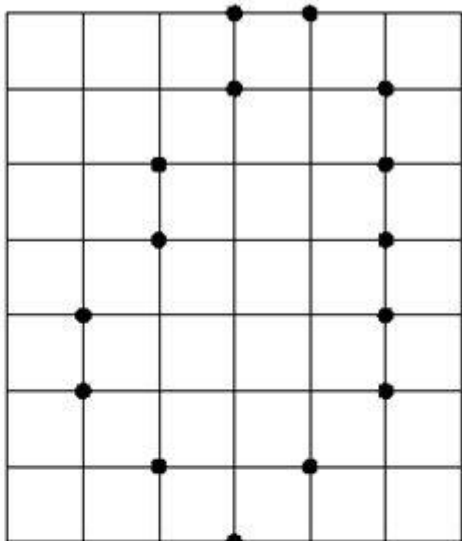
After resampling: 076666...12

Examples of Chain Codes

Object boundary (resampling)

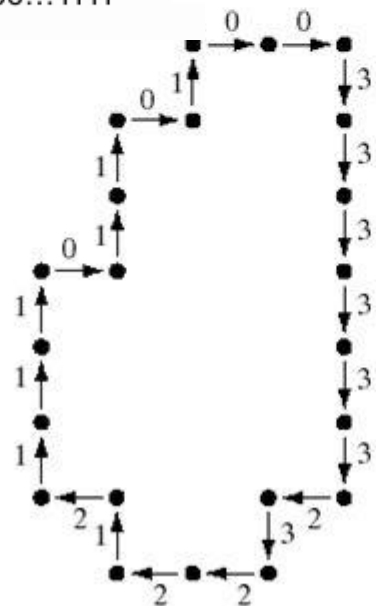


Boundary vertices

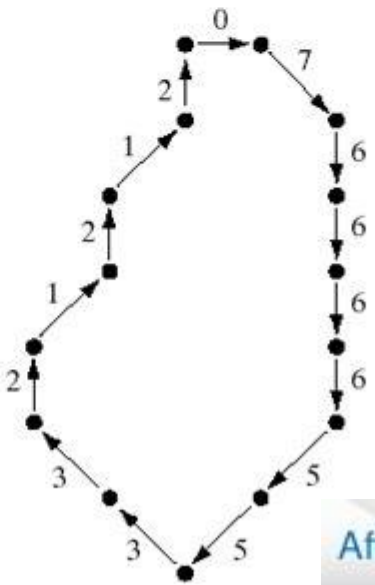


Before resample: 000000766...1111

4-directional chain code



8-directional chain code



After resampling: 076666...12

Problem2:

- start point

Solution:

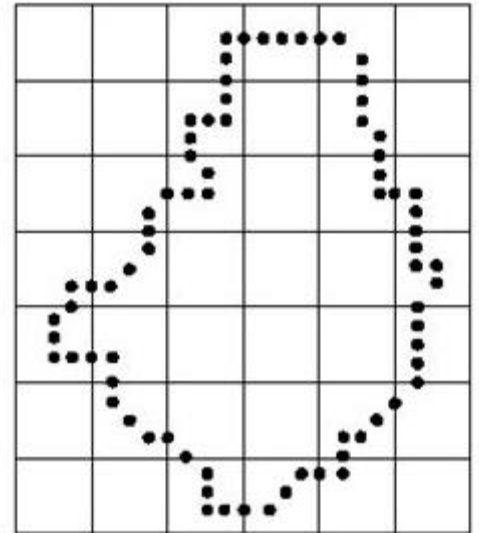
- Normalizations

Disadvantages

- Long chains of codes
- No invariance to Rotation and Scale
- Sensitive to Noise

Solution:

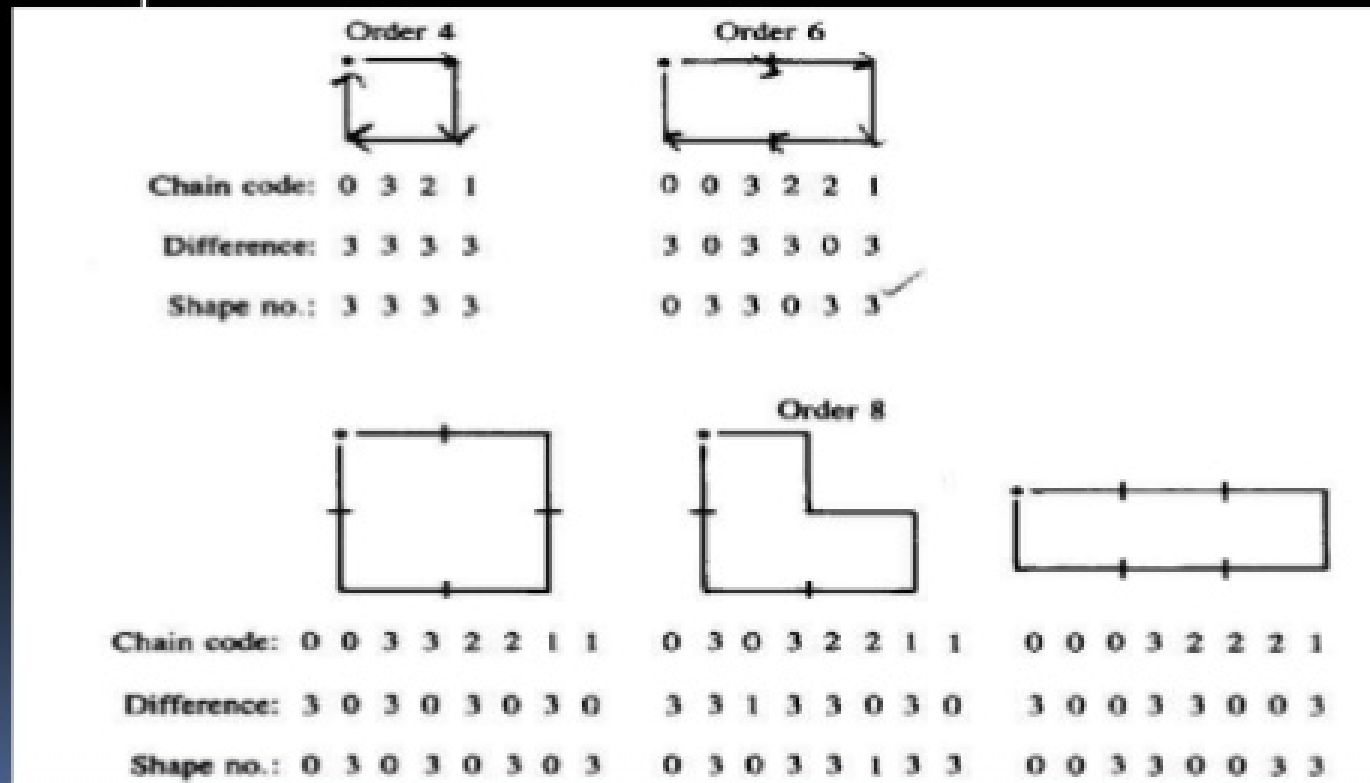
- Re-sample the image to a lower resolution before calculating the code



- **Shape number**

- compute the chain code difference
- re-order this to create the minimum integer
- this is called the shape number

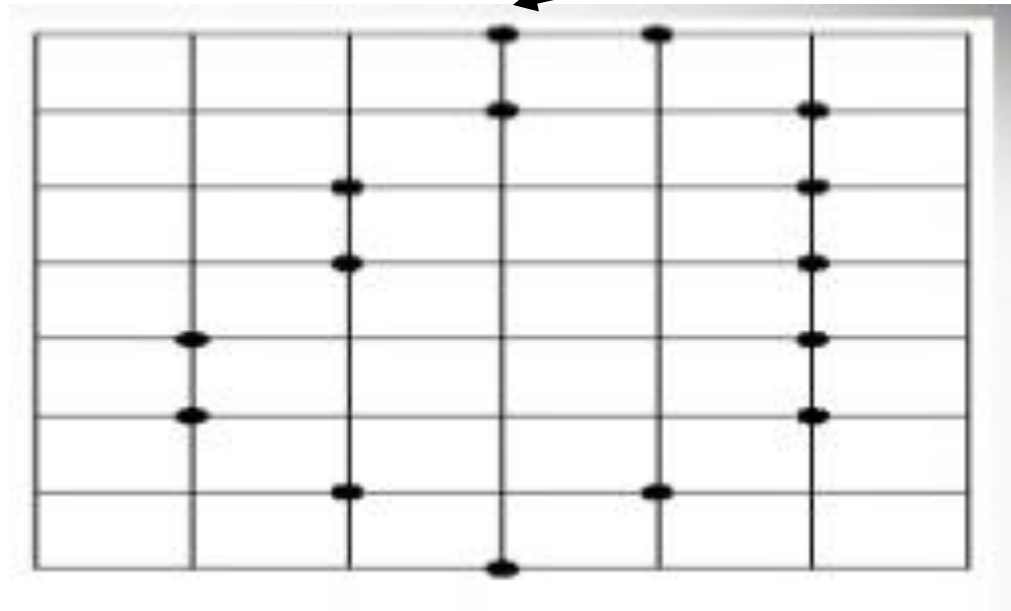
- **Example:-**



Class Work

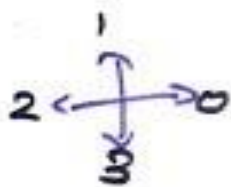
- An image is given as in the figure below. Write down the 8-chain code and find the shape number of it.

Starting point

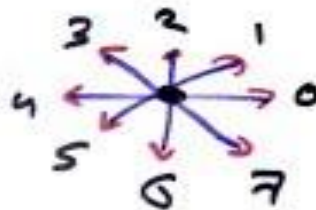


<https://youtu.be/NPyBHVmlyDY>

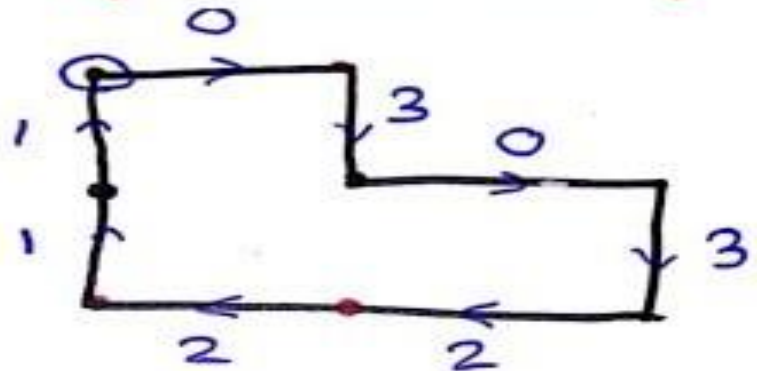
Chain Code (Hindi)



4-Point



8 Point



Normalize Chain Code

<https://youtu.be/Dd3xWUi7JIY>

